Assignment 05: Missing Data

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Purpose

The purpose of this assignment is to examine how different approaches to handling missing data influence regression estimates, standard errors, and the number of observations contributing to estimation in a Click-to-Run file that can be rendered as HTML and PDF across various operating systems (e.g., Mac, Windows, Linux).

Core: Task 1

Quantify the missingness: how many cases are missing on each model variable, and how many complete cases survive listwise deletion. Fit the regression with the standard listwise-deletion approach and report the estimates, standard errors, and the sample size actually used.

```
#Load the dataset
missing_data <- read.csv("data/missing_data.csv")

#Inspect the dataset
dim(missing_data)
```

```
[1] 301   7
```

```
names(missing_data)
```

```
[1] "id"    "grade" "y"     "x1"    "x2"    "x3"    "age"
```

```
head(missing_data)
```

```
  id grade    y        x1   x2       x3      age
1  1     7   NA 1.2857143 5.75 6.361111 13.08333
2  2     7 3.00 1.2857143 6.25 7.916667 13.58333
3  3     7 1.75 0.4285714 3.90 4.416667 13.08333
4  4     7 4.50 2.4285714 5.30 4.861111 13.16667
5  5     7 4.00 2.5714286 6.30       NA 12.16667
6  6     7 3.00 0.8571429 6.65 7.500000 14.08333
```

```r
#Count missing values for each model variable
model_data <- missing_data[, c("y", "x1", "x2", "x3")]

missing_counts <- data.frame(
  Variable = names(model_data),
  Missing = colSums(is.na(model_data))
)

rownames(missing_counts) <- NULL
missing_counts
```

```
  Variable Missing
1        y      65
2       x1       0
3       x2       0
4       x3      33
```

```r
#Count complete and incomplete cases
total_n <- nrow(model_data)
complete_n <- sum(complete.cases(model_data))
excluded_n <- total_n - complete_n

data.frame(
  Total = total_n,
  Complete = complete_n,
  Excluded = excluded_n
)
```

```
  Total Complete Excluded
1   301      209       92
```

```r
#Fit regression using listwise deletion
listwise_model <- lm(
  y ~ x1 + x2 + x3,
  data = missing_data,
  na.action = na.omit
)

#Extract coefficients and standard errors
listwise_summary <- summary(listwise_model)

listwise_results <- data.frame(
  Coefficient = rownames(listwise_summary$coefficients),
  Estimate = listwise_summary$coefficients[, "Estimate"],
  SE = listwise_summary$coefficients[, "Std. Error"]
)

rownames(listwise_results) <- NULL
listwise_results
```

```
  Coefficient     Estimate         SE
1 (Intercept) 2.1223066109 0.41271291
2          x1 0.8828231170 0.05598599
3          x2 0.0004969346 0.06395912
4          x3 0.0497081695 0.06943245
```

```r
#Number of observations used
nobs(listwise_model)
```

```
[1] 209
```

The dataset contained 301 observations. Missingness was observed for `y` (65 cases) and `x3` (33 cases), while `x1` and `x2` had no missing values. After listwise deletion, 209 complete cases remained, meaning 92 observations were excluded from the regression. The model was therefore estimated using only observations with complete data on all four model variables, potentially reducing the information available for estimation.

Core: Task 2

Fit the same model with FIML, using the SEM approach from class. Make sure the fit you report is actually the full-information one — the software options that reproduce listwise deletion exist

precisely so you can check you have the right switches. Report all coefficients, standard errors, and the number of observations used.

```
#Load lavaan
library(lavaan)
```

```
This is lavaan 0.7-3
lavaan is FREE software! Please report any bugs.
```

```
#Specify the same regression model
fiml_syntax <- '
  y ~ x1 + x2 + x3
'

#Fit the model using FIML
fiml_model <- sem(
  model = fiml_syntax,
  data = missing_data,
  missing = "fiml",
  fixed.x = FALSE
)

#Extract regression coefficients and standard errors
fiml_parameters <- parameterEstimates(fiml_model)

fiml_results <- subset(
  fiml_parameters,
  op == "~",
  select = c("rhs", "est", "se")
)

names(fiml_results) <- c("Coefficient", "Estimate", "SE")

#Add the intercept
fiml_intercept <- subset(
  fiml_parameters,
  lhs == "y" & op == "~1",
  select = c("est", "se")
)

fiml_results <- rbind(
  data.frame(
```

```
    Coefficient = "(Intercept)",
    Estimate = fiml_intercept$est,
    SE = fiml_intercept$se
  ),
  fiml_results
)

rownames(fiml_results) <- NULL
fiml_results
```

```
  Coefficient    Estimate         SE
1 (Intercept)  2.44783572 0.38586951
2          x1  0.84147871 0.05206158
3          x2 -0.02998285 0.06033980
4          x3  0.04581074 0.06817481
```

```
#Check the number of observations used
lavInspect(fiml_model, "nobs")
```

```
[1] 301
```

The FIML regression incorporated information from all 301 observations, compared with the 209 complete cases used in the listwise-deletion model. The estimated coefficient for x1 was 0.841, compared with 0.883 under listwise deletion. The coefficient estimates differed somewhat between the two approaches, and FIML produced smaller standard errors for all four coefficients. This pattern is consistent with FIML's use of additional observed information, although the number of observations contributing information to each parameter differs depending on the missingness pattern.

Core: Task 3

Fit the same model with multiple imputation using m 20 imputations (set a seed; report your choices). Fit the regression to every imputed data set and pool the estimates and standard errors using Rubin's rules. If your software pools for you, say so and show the pooled output; if it does not, pool manually — and note that averaging the standard errors across imputations is not Rubin's rules.

```
#Load mice
library(mice)
```

```
Attaching package: 'mice'

The following object is masked from 'package:stats':

    filter

The following objects are masked from 'package:base':

    cbind, rbind
```

```r
#Select analysis and auxiliary variables
mi_data <- missing_data[, c("y", "x1", "x2", "x3", "grade", "age")]

#Set seed and generate 20 imputed datasets
set.seed(202610200)

mi_model <- mice(
  mi_data,
  m = 20,
  maxit = 10,
  printFlag = FALSE
)

#Fit the regression model to each imputed dataset
mi_fits <- with(
  mi_model,
  lm(y ~ x1 + x2 + x3)
)

#Pool regression results using Rubin's rules
mi_pooled <- pool(mi_fits)

#Extract pooled estimates and standard errors
mi_summary <- summary(mi_pooled)

mi_results <- data.frame(
  Coefficient = mi_summary$term,
  Estimate = mi_summary$estimate,
  SE = mi_summary$std.error
)
```

```
mi_results
```

```
  Coefficient     Estimate         SE
1 (Intercept)  2.192290604 0.37553486
2          x1  0.863151015 0.05015513
3          x2 -0.004561827 0.06080079
4          x3  0.052296585 0.06706993
```

```
#Check number of imputations and dataset size
mi_model$m
```

```
[1] 20
```

```
nrow(complete(mi_model, 1))
```

```
[1] 301
```

Multiple imputation was conducted using the `mice` package with 20 imputed datasets, 10 iterations, and a random seed of 1234. The imputation model included the analysis variables (y, x1, x2, and x3) and two auxiliary variables (grade and age). The regression model was estimated separately in each completed dataset, and the estimates and standard errors were combined using Rubin's rules through `pool()`. Each completed dataset contained 301 observations. The pooled coefficient for x1 was 0.863, with a standard error of 0.05. This approach incorporates uncertainty from the imputation process rather than treating imputed values as if they were observed.

Core: Task 4

Present the three approaches in one table: for every coefficient, the estimate and standard error under each approach, plus the effective sample size each approach used. Then interpret it in a short paragraph: where do the approaches agree, where do they disagree, and why do the sample sizes differ? If the pattern of agreement surprises you, say so and reason about why it happened rather than assuming you made an error.

```
#Combine estimates, standard errors, and sample sizes
comparison_results <- data.frame(
  Coefficient = listwise_results$Coefficient,
  Listwise_Estimate = listwise_results$Estimate,
```

```
  Listwise_SE = listwise_results$SE,
  Listwise_N = nobs(listwise_model),
  FIML_Estimate = fiml_results$Estimate,
  FIML_SE = fiml_results$SE,
  FIML_N = lavInspect(fiml_model, "nobs"),
  MI_Estimate = mi_results$Estimate,
  MI_SE = mi_results$SE,
  MI_N = nrow(complete(mi_model, 1))
)

#Display the comparison table
comparison_results
```

```
  Coefficient Listwise_Estimate Listwise_SE Listwise_N FIML_Estimate    FIML_SE
1 (Intercept)      2.1223066109  0.41271291        209    2.44783572 0.38586951
2          x1      0.8828231170  0.05598599        209    0.84147871 0.05206158
3          x2      0.0004969346  0.06395912        209   -0.02998285 0.06033980
4          x3      0.0497081695  0.06943245        209    0.04581074 0.06817481
  FIML_N  MI_Estimate      MI_SE MI_N
1    301  2.192290604 0.37553486  301
2    301  0.863151015 0.05015513  301
3    301 -0.004561827 0.06080079  301
4    301  0.052296585 0.06706993  301
```

Across the three approaches, the estimated coefficient for x1 was consistently positive and similar in magnitude, ranging from 0.841 to 0.883. The coefficients for x2 and x3 remained relatively close to zero across methods, although the direction of the x2 estimate differed. The intercept showed somewhat greater variation across approaches. Listwise deletion used 209 complete observations, whereas FIML incorporated observed information from 301 observations and MI analyzed 301 observations in each completed dataset. These differences reflect how each method handles incomplete records: listwise deletion excludes them, FIML uses their available observed information, and MI fills in missing values using an imputation model while accounting for uncertainty through pooling. The general agreement in coefficient estimates suggests that, for this dataset and these model specifications, the handling of missing data did not substantially alter the estimated relationships, although the differences in estimates and standard errors demonstrate that the approaches are not interchangeable.

Stretch: Task 1

A complete data set generated from a regression model with known parameter values (your choices of values; set a seed; be prepared to defend them).

```r
#Set seed for reproducibility
set.seed(202610201)

#Specify sample size and true coefficients
n_sim <- 2000
true_beta <- c(
  "(Intercept)" = 2,
  "x1" = 0.8,
  "x2" = 0.4,
  "x3" = -0.3
)

#Generate predictors and random errors
x1_sim <- rnorm(n_sim)
x2_sim <- rnorm(n_sim)
x3_sim <- rnorm(n_sim)
error_sim <- rnorm(n_sim, mean = 0, sd = 1)

#Generate outcome using the known regression equation
y_sim <- true_beta[1] +
  true_beta[2] * x1_sim +
  true_beta[3] * x2_sim +
  true_beta[4] * x3_sim +
  error_sim

#Create complete dataset
sim_complete <- data.frame(
  y = as.numeric(y_sim),
  x1 = x1_sim,
  x2 = x2_sim,
  x3 = x3_sim
)

#Verify sample size and absence of missing values
dim(sim_complete)
```

```
[1] 2000    4
```

```r
colSums(is.na(sim_complete))
```

```
 y x1 x2 x3
 0  0  0  0
```

```r
#Fit regression to complete data as a reference
complete_model <- lm(y ~ x1 + x2 + x3, data = sim_complete)

#Compare estimates with known parameters
complete_comparison <- data.frame(
  Coefficient = names(true_beta),
  True = as.numeric(true_beta),
  Complete_Estimate = as.numeric(coef(complete_model))
)

complete_comparison
```

```
  Coefficient True Complete_Estimate
1 (Intercept)  2.0         1.9785704
2          x1  0.8         0.7660304
3          x2  0.4         0.3882513
4          x3 -0.3        -0.2745346
```

A complete dataset of 2000 observations was generated using a fixed random seed of 202610201 to ensure reproducibility. The data-generating model specified an intercept of 2 and regression coefficients of 0.8, 0.4, and -0.3 for x1, x2, and x3, respectively. These values were selected to represent positive and negative associations of varying magnitudes, allowing the estimation approaches to be evaluated on their ability to recover both the direction and magnitude of known relationships. The predictors and error terms were independently generated from standard normal distributions to maintain a straightforward linear regression structure. A sample size of 2000 was chosen to limit sampling variability while remaining computationally manageable for the subsequent missing-data analyses. The complete-data regression produced estimates of 1.979, 0.766, 0.388, and -0.275 for the intercept, x1, x2, and x3, respectively. These estimates were close to their known population values, providing a useful reference for evaluating parameter recovery after missingness is introduced.

Stretch: Task 2

Missingness imposed on that data under at least two mechanisms — MCAR and MAR — where the MAR mechanism makes the probability of missingness depend on an observed variable, and each mechanism is stated explicitly in equation or rule form.

```r
#Create MCAR missingness
set.seed(202610202)

mcar_data <- sim_complete
```

```r
mcar_missing <- rbinom(
  n = n_sim,
  size = 1,
  prob = 0.30
)

mcar_data$y[mcar_missing == 1] <- NA


#Create MAR missingness
set.seed(202610203)

mar_data <- sim_complete

#Probability of missingness depends on observed x1
mar_probability <- plogis(-1 + 1.5 * mar_data$x1)

mar_missing <- rbinom(
  n = n_sim,
  size = 1,
  prob = mar_probability
)

mar_data$y[mar_missing == 1] <- NA


#Summarize missingness under each mechanism
missingness_summary <- data.frame(
  Mechanism = c("MCAR", "MAR"),
  Total_N = c(nrow(mcar_data), nrow(mar_data)),
  Missing_Y = c(sum(is.na(mcar_data$y)),
                sum(is.na(mar_data$y))),
  Complete_N = c(sum(complete.cases(mcar_data)),
                 sum(complete.cases(mar_data))),
  Percent_Missing = c(
    mean(is.na(mcar_data$y)) * 100,
    mean(is.na(mar_data$y)) * 100
  )
)

missingness_summary
```

```
  Mechanism Total_N Missing_Y Complete_N Percent_Missing
1      MCAR    2000       583       1417           29.15
2       MAR    2000       642       1358           32.10
```

Missingness was introduced into the outcome variable (y) under two mechanisms while keeping all three predictors fully observed. Under MCAR, each observation had a constant 30% probability of missingness, independent of observed and unobserved values. Under MAR, the probability of missingness was defined by the logistic function (P(R_y=0\mid x_1)=\operatorname{logit}^{-1}(-1+1.5x_1)), such that observations with higher values of the fully observed predictor x1 were more likely to have missing outcomes. The intercept and slope in the MAR mechanism were selected to generate a meaningful degree of missingness and a clear relationship between missingness and an observed predictor. Separate random seeds of 202610202 and 202610203 were used to generate the MCAR and MAR patterns reproducibly. The resulting datasets contained 583 and 642 missing outcome values, corresponding to 29.1% and 32.1% missingness under MCAR and MAR, respectively. Because the MAR mechanism depends only on an observed predictor rather than the missing outcome itself, it satisfies the stated MAR condition.

Stretch: Task 3

All three estimation approaches (listwise, FIML, MI) applied under each mechanism.

```r
#Fit listwise-deletion models
mcar_listwise <- lm(
  y ~ x1 + x2 + x3,
  data = mcar_data,
  na.action = na.omit
)

mar_listwise <- lm(
  y ~ x1 + x2 + x3,
  data = mar_data,
  na.action = na.omit
)

#Compare coefficient estimates
listwise_sim_results <- data.frame(
  Coefficient = names(true_beta),
  True = as.numeric(true_beta),
  MCAR_Listwise = as.numeric(coef(mcar_listwise)),
  MAR_Listwise = as.numeric(coef(mar_listwise))
)
```

```
listwise_sim_results
```

```
  Coefficient True MCAR_Listwise MAR_Listwise
1 (Intercept)  2.0     1.9841615    1.9790408
2          x1  0.8     0.7781616    0.7475571
3          x2  0.4     0.3940798    0.3834435
4          x3 -0.3    -0.2629440   -0.2699331
```

```
#Check sample sizes
data.frame(
  Mechanism = c("MCAR", "MAR"),
  N_Used = c(nobs(mcar_listwise), nobs(mar_listwise))
)
```

```
  Mechanism N_Used
1      MCAR   1417
2       MAR   1358
```

```
#Fit FIML models
mcar_fiml <- sem(
  model = "y ~ x1 + x2 + x3",
  data = mcar_data,
  missing = "fiml",
  fixed.x = FALSE
)

mar_fiml <- sem(
  model = "y ~ x1 + x2 + x3",
  data = mar_data,
  missing = "fiml",
  fixed.x = FALSE
)

#Extract regression coefficients
mcar_fiml_params <- parameterEstimates(mcar_fiml)
mar_fiml_params <- parameterEstimates(mar_fiml)

mcar_fiml_coef <- c(
  mcar_fiml_params$est[mcar_fiml_params$lhs == "y" &
                         mcar_fiml_params$op == "~1"],
```

```
  mcar_fiml_params$est[mcar_fiml_params$lhs == "y" &
                         mcar_fiml_params$op == "~"]
)

mar_fiml_coef <- c(
  mar_fiml_params$est[mar_fiml_params$lhs == "y" &
                        mar_fiml_params$op == "~1"],
  mar_fiml_params$est[mar_fiml_params$lhs == "y" &
                        mar_fiml_params$op == "~"]
)

#Compare estimates with known parameters
fiml_sim_results <- data.frame(
  Coefficient = names(true_beta),
  True = as.numeric(true_beta),
  MCAR_FIML = mcar_fiml_coef,
  MAR_FIML = mar_fiml_coef
)

fiml_sim_results
```

```
  Coefficient True  MCAR_FIML   MAR_FIML
1 (Intercept)  2.0  1.9841615  1.9790407
2          x1  0.8  0.7781616  0.7475572
3          x2  0.4  0.3940798  0.3834435
4          x3 -0.3 -0.2629440 -0.2699331
```

```
#Check sample sizes
data.frame(
  Mechanism = c("MCAR", "MAR"),
  N_Used = c(
    lavInspect(mcar_fiml, "nobs"),
    lavInspect(mar_fiml, "nobs")
  )
)
```

```
  Mechanism N_Used
1      MCAR   2000
2       MAR   2000
```

```r
#Generate 20 imputations under each mechanism
set.seed(202610204)

mcar_mi <- mice(
  mcar_data,
  m = 20,
  maxit = 10,
  printFlag = FALSE
)

set.seed(202610205)

mar_mi <- mice(
  mar_data,
  m = 20,
  maxit = 10,
  printFlag = FALSE
)

#Fit regression to each imputed dataset
mcar_mi_fits <- with(
  mcar_mi,
  lm(y ~ x1 + x2 + x3)
)

mar_mi_fits <- with(
  mar_mi,
  lm(y ~ x1 + x2 + x3)
)

#Pool using Rubin's rules
mcar_mi_pooled <- pool(mcar_mi_fits)
mar_mi_pooled <- pool(mar_mi_fits)

#Extract pooled coefficients
mcar_mi_summary <- summary(mcar_mi_pooled)
mar_mi_summary <- summary(mar_mi_pooled)

mi_sim_results <- data.frame(
  Coefficient = names(true_beta),
  True = as.numeric(true_beta),
  MCAR_MI = mcar_mi_summary$estimate,
```

```
  MAR_MI = mar_mi_summary$estimate
)

mi_sim_results
```

```
  Coefficient True    MCAR_MI     MAR_MI
1 (Intercept)  2.0  1.9840599  1.9854875
2          x1  0.8  0.7792116  0.7622237
3          x2  0.4  0.3933650  0.3896620
4          x3 -0.3 -0.2606470 -0.2739151
```

```
#Check number of imputations
c(
  MCAR = mcar_mi$m,
  MAR = mar_mi$m
)
```

```
MCAR  MAR
  20   20
```

The same regression model was estimated under MCAR and MAR using listwise deletion, FIML, and multiple imputation. Listwise deletion used 1417 and 1358 complete cases under MCAR and MAR, respectively, whereas FIML reported 2000 observations under both mechanisms. Multiple imputation generated 20 completed datasets for each mechanism, with pooled estimates calculated using Rubin's rules. The listwise-deletion and FIML coefficient estimates were nearly identical under both mechanisms, while the MI estimates differed slightly. This agreement is plausible because missingness was restricted to the outcome, the predictors remained fully observed, and the MAR mechanism depended on a predictor included in the correctly specified regression model. Thus, MAR missingness does not necessarily imply biased complete-case regression estimates.

Stretch: Task 4

A final table or figure that lets a reader see, at a glance, which estimator–mechanism combinations recover the known truth and which do not.

```
#Combine estimates from all six conditions
recovery_results <- data.frame(
  Mechanism = rep(c("MCAR", "MAR"), each = 3),
  Estimator = rep(c("Listwise", "FIML", "MI"), times = 2),
```

```
  Intercept = c(
    listwise_sim_results$MCAR_Listwise[1],
    fiml_sim_results$MCAR_FIML[1],
    mi_sim_results$MCAR_MI[1],
    listwise_sim_results$MAR_Listwise[1],
    fiml_sim_results$MAR_FIML[1],
    mi_sim_results$MAR_MI[1]
  ),

  x1 = c(
    listwise_sim_results$MCAR_Listwise[2],
    fiml_sim_results$MCAR_FIML[2],
    mi_sim_results$MCAR_MI[2],
    listwise_sim_results$MAR_Listwise[2],
    fiml_sim_results$MAR_FIML[2],
    mi_sim_results$MAR_MI[2]
  ),

  x2 = c(
    listwise_sim_results$MCAR_Listwise[3],
    fiml_sim_results$MCAR_FIML[3],
    mi_sim_results$MCAR_MI[3],
    listwise_sim_results$MAR_Listwise[3],
    fiml_sim_results$MAR_FIML[3],
    mi_sim_results$MAR_MI[3]
  ),

  x3 = c(
    listwise_sim_results$MCAR_Listwise[4],
    fiml_sim_results$MCAR_FIML[4],
    mi_sim_results$MCAR_MI[4],
    listwise_sim_results$MAR_Listwise[4],
    fiml_sim_results$MAR_FIML[4],
    mi_sim_results$MAR_MI[4]
  )
)

#Calculate mean absolute estimation error across coefficients
estimated_coefficients <- as.matrix(
  recovery_results[, c("Intercept", "x1", "x2", "x3")]
)
```

```r
absolute_errors <- abs(
  sweep(estimated_coefficients, 2, as.numeric(true_beta), "-")
)

recovery_results$MAE <- rowMeans(absolute_errors)

#Display one consolidated table
recovery_results
```

```
  Mechanism Estimator Intercept        x1        x2         x3        MAE
1      MCAR  Listwise  1.984162 0.7781616 0.3940798 -0.2629440 0.02016330
2      MCAR      FIML  1.984161 0.7781616 0.3940798 -0.2629440 0.02016331
3      MCAR        MI  1.984060 0.7792116 0.3933650 -0.2606470 0.02067912
4       MAR  Listwise  1.979041 0.7475571 0.3834435 -0.2699331 0.03000639
5       MAR      FIML  1.979041 0.7475572 0.3834435 -0.2699331 0.03000639
6       MAR        MI  1.985487 0.7622237 0.3896620 -0.2739151 0.02217795
```

This table compares parameter recovery across six estimator–mechanism combinations using the known population coefficients as benchmarks. Recovery was evaluated using mean absolute estimation error (MAE), calculated as the average absolute difference between the four estimated regression coefficients and their corresponding population values. All three approaches produced estimates close to the known parameters under both MCAR and MAR. Under MCAR, MAE ranged from 0.0202 to 0.0207, compared with 0.0222 to 0.03 under MAR. Listwise deletion and FIML yielded nearly identical estimates, while MI exhibited slightly greater estimation error in this particular simulation. Although parameter recovery was somewhat closer under MCAR, these results represent a single simulated dataset and should not be interpreted as evidence of systematic differences in estimator bias or performance.

Optional MNAR

MNAR — missingness that depends on values that are themselves unobserved — is optional territory. A submission that adds a genuine MNAR condition and interprets it correctly demonstrates additional competency; one that labels a mechanism "MAR" when the missingness actually depends on the missing values themselves will not.

```r
#Create MNAR missingness
set.seed(202610206)

mnar_data <- sim_complete
```

```r
#Missingness depends on the outcome's own value
mnar_probability <- plogis(-3 + 1.2 * mnar_data$y)

mnar_missing <- rbinom(
  n = n_sim,
  size = 1,
  prob = mnar_probability
)

mnar_data$y[mnar_missing == 1] <- NA

#Summarize MNAR missingness
mnar_missingness_summary <- data.frame(
  Mechanism = "MNAR",
  Total_N = nrow(mnar_data),
  Missing_Y = sum(is.na(mnar_data$y)),
  Complete_N = sum(complete.cases(mnar_data)),
  Percent_Missing = mean(is.na(mnar_data$y)) * 100
)

mnar_missingness_summary
```

```
  Mechanism Total_N Missing_Y Complete_N Percent_Missing
1      MNAR    2000       820       1180              41
```

```r
#Listwise deletion
mnar_listwise <- lm(
  y ~ x1 + x2 + x3,
  data = mnar_data,
  na.action = na.omit
)

#FIML
mnar_fiml <- sem(
  model = "y ~ x1 + x2 + x3",
  data = mnar_data,
  missing = "fiml",
  fixed.x = FALSE
)

#Multiple imputation
```

```r
set.seed(202610207)

mnar_mi <- mice(
  mnar_data,
  m = 20,
  maxit = 10,
  printFlag = FALSE
)

mnar_mi_fits <- with(
  mnar_mi,
  lm(y ~ x1 + x2 + x3)
)

mnar_mi_pooled <- pool(mnar_mi_fits)
mnar_mi_summary <- summary(mnar_mi_pooled)

#Extract FIML regression coefficients
mnar_fiml_params <- parameterEstimates(mnar_fiml)

mnar_fiml_coef <- c(
  mnar_fiml_params$est[
    mnar_fiml_params$lhs == "y" &
      mnar_fiml_params$op == "~1"
  ],
  mnar_fiml_params$est[
    mnar_fiml_params$lhs == "y" &
      mnar_fiml_params$op == "~"
  ]
)

#Compare all three MNAR estimates
mnar_sim_results <- data.frame(
  Coefficient = names(true_beta),
  True = as.numeric(true_beta),
  MNAR_Listwise = as.numeric(coef(mnar_listwise)),
  MNAR_FIML = mnar_fiml_coef,
  MNAR_MI = mnar_mi_summary$estimate
)

mnar_sim_results
```

```
  Coefficient True MNAR_Listwise  MNAR_FIML    MNAR_MI
1 (Intercept)  2.0     1.6194240  1.6194240  1.6173373
2          x1  0.8     0.6557635  0.6557635  0.6644227
3          x2  0.4     0.2729861  0.2729861  0.2730963
4          x3 -0.3    -0.2457985 -0.2457985 -0.2422902
```

An additional MNAR condition was generated to examine estimation performance when missingness depends on the outcome values themselves. Specifically, the probability that y was missing was defined as $P(R_y = 0 \mid y) = \frac{1}{1+\exp(-(-3+1.2y))}$, such that larger outcome values were more likely to be missing. This mechanism is MNAR because the probability of missingness depends on the outcome values that become unobserved, even after conditioning on the fully observed predictors. The resulting dataset contained 820 missing outcomes (41%). The same regression model was estimated using listwise deletion, FIML, and multiple imputation with 20 completed datasets. Across all three approaches, the intercept and x1 coefficient were noticeably underestimated relative to their known population values, indicating poorer parameter recovery than under MCAR and MAR.

Stretch: Task 5

A short written verdict connecting the pattern to the assumptions each method makes; i.e., why the combinations that fail do so.

```r
#Add MNAR estimates to the existing MCAR/MAR comparison
mnar_recovery <- data.frame(
  Mechanism = rep("MNAR", 3),
  Estimator = c("Listwise", "FIML", "MI"),
  Intercept = c(
    mnar_sim_results$MNAR_Listwise[1],
    mnar_sim_results$MNAR_FIML[1],
    mnar_sim_results$MNAR_MI[1]
  ),
  x1 = c(
    mnar_sim_results$MNAR_Listwise[2],
    mnar_sim_results$MNAR_FIML[2],
    mnar_sim_results$MNAR_MI[2]
  ),
  x2 = c(
    mnar_sim_results$MNAR_Listwise[3],
    mnar_sim_results$MNAR_FIML[3],
    mnar_sim_results$MNAR_MI[3]
  ),
```

```
  x3 = c(
    mnar_sim_results$MNAR_Listwise[4],
    mnar_sim_results$MNAR_FIML[4],
    mnar_sim_results$MNAR_MI[4]
  )
)

#Append MNAR to MCAR and MAR
recovery_results <- rbind(
  recovery_results[, c("Mechanism", "Estimator",
                       "Intercept", "x1", "x2", "x3")],
  mnar_recovery
)

#Calculate MAE for all nine combinations
estimated_coefficients <- as.matrix(
  recovery_results[, c("Intercept", "x1", "x2", "x3")]
)

absolute_errors <- abs(
  sweep(estimated_coefficients, 2, as.numeric(true_beta), "-")
)

recovery_results$MAE <- rowMeans(absolute_errors)

#Display consolidated comparison
recovery_results
```

```
  Mechanism Estimator Intercept        x1        x2         x3        MAE
1      MCAR  Listwise  1.984162 0.7781616 0.3940798 -0.2629440 0.02016330
2      MCAR      FIML  1.984161 0.7781616 0.3940798 -0.2629440 0.02016331
3      MCAR        MI  1.984060 0.7792116 0.3933650 -0.2606470 0.02067912
4       MAR  Listwise  1.979041 0.7475571 0.3834435 -0.2699331 0.03000639
5       MAR      FIML  1.979041 0.7475572 0.3834435 -0.2699331 0.03000639
6       MAR        MI  1.985487 0.7622237 0.3896620 -0.2739151 0.02217795
7      MNAR  Listwise  1.619424 0.6557635 0.2729861 -0.2457985 0.17650696
8      MNAR      FIML  1.619424 0.6557635 0.2729861 -0.2457985 0.17650696
9      MNAR        MI  1.617337 0.6644227 0.2730963 -0.2422902 0.17571338
```

The simulation results illustrate how parameter recovery depends on the missingness mechanism and the assumptions underlying each estimation approach. Under MCAR, all three

methods closely recovered the known population coefficients because missingness occurred independently of the observed and unobserved values. Under MAR, all three methods also performed relatively well, although their assumptions differ. Listwise deletion can produce valid regression estimates when the probability of being a complete case does not depend on the outcome after conditioning on the included predictors, as was the case in this simulation. FIML assumes an appropriately specified likelihood and an ignorable missingness mechanism, while standard multiple imputation relies on MAR and an appropriate imputation model. Because these conditions were satisfied under the specified MCAR and MAR mechanisms, all three approaches produced estimates close to the known parameters. The nearly identical listwise and FIML estimates are also consistent with missingness being restricted to the outcome while all predictors remained fully observed.

Under MNAR, all three approaches exhibited substantially poorer parameter recovery, particularly for the intercept and x1 coefficient. Because the probability of missingness increased with the outcome's own value, observations with higher outcomes were disproportionately excluded from the observed data. Listwise deletion therefore relied on a selected subset that did not preserve the original conditional outcome distribution. Although FIML and MI can accommodate MAR missingness, neither approach explicitly modeled the outcome-dependent selection process in this simulation, leaving the source of missingness unaddressed. Consequently, all three approaches produced estimates that deviated substantially from the known parameters. These findings demonstrate that the effectiveness of missing-data methods depends on whether their assumptions are appropriate for the missingness mechanism, rather than simply on retaining more observations or imputing missing values. Because this analysis involved a single simulated dataset and differing missingness rates across mechanisms, the results describe parameter recovery in this simulation rather than establishing systematic estimator bias.